

The composite-dimensional Heisenberg–Weyl Bohnenblust–Hille question

Literature-already-answered note for arXiv:2301.01438

11 August 2026

Status. Answered in the direct same-author split/follow-up arXiv:2406.08509, with a degree loss. The answer gives a dimension-free inequality for every composite local dimension. It does not establish the prime-case exponent and does not settle optimal growth of the quantum BH constants.

Source question

The conclusion of Klein–Slote–Volberg–Zhang [1], on source PDF page 32, says that the eigenspace relations needed for qudit Bohnenblust–Hille inequalities are mostly complete, but asks what can be said for the Heisenberg–Weyl (HW) basis when K is composite. It then asks the separate question whether exponential dependence of the quantum BH constants on degree is necessary.

Direct later answer

Slote–Volberg–Zhang [2] is the focused qudit-BH paper by three of the four source authors. Its arXiv record explains that an old version had appeared as arXiv:2301.01438v2 before that record was replaced, and that the focused version extends the results to more general qudit systems. This makes the source relation direct rather than inferential.

Fix a non-prime integer $K \geq 4$, $d \geq 1$, and an observable $A \in M_K(\mathbb{C})^{\otimes n}$ of HW degree at most d , using the degree convention of [2]. Write $\hat{A}$ for its HW coefficient array. Theorem 7 of [2], on supporting PDF pages 5–6, proves

$$\left\| \hat{A} \right\|_{\frac{2(K-1)d}{(K-1)d+1}} \leq C(d, K) \|A\|_{\text{op}}, \quad C(d, K) \leq K^{2d} \text{BH}_{\Omega_K}^{\leq (K-1)d}. \quad (1)$$

Moreover, K^{2d} may be replaced by $|\Sigma_K|^d$, where

$$\Sigma_K = \{(\ell, m) \in \mathbb{Z}_K^2 : \ell \text{ and } m \text{ are coprime}\}.$$

The cyclic-group BH constant on the right is finite independently of n . Thus (1) is genuinely dimension-free in the tensor length.

Idea of the proof in the answering paper

For prime K , the nonzero cyclic subgroups of $\mathbb{Z}_K^2$ form a clean partition. For composite K , primitive-generated subgroups overlap, and an HW element can have spectrum either Ω_K or the shifted root

set $\Omega_{2K} \setminus \Omega_K$. The later paper accommodates both effects by averaging suitable eigenprojections into density matrices $\rho(\omega)$.

The scalar function

$$f_A(\omega) = \text{tr}(A\rho(\omega))$$

has degree at most $(K-1)d$. Distinct HW profiles contribute disjoint sets of scalar monomials, so their coefficients cannot cancel across profiles; the averaging costs at most a factor $|\Sigma_K|^d$. The cyclic-group BH inequality at degree $(K-1)d$, together with $|\text{tr}(A\rho)| \leq \|A\|_{\text{op}}$, yields (1).

Exact scope

The source asked broadly what can be said in composite local dimension, and (1) supplies a uniform theorem for every non-prime $K \geq 4$. So that residual is literature-already-answered. The modification is substantive: the prime-case scalar degree d becomes $(K-1)d$, and the coefficient exponent changes accordingly from $2d/(d+1)$ to $2(K-1)d/((K-1)d+1)$. This note therefore makes no claim that the prime-strength composite inequality holds. Nor does [2] settle the source's separate optimal-growth question about whether exponential dependence on d is necessary.

Provenance. This is a literature-status identification, not new mathematics from the run. Exact arXiv-id and core-keyword searches through 11 August 2026 found no prior packet recording this precise source-to-answer match.

References

- [1] O. Klein, J. Slote, A. Volberg, and H. Zhang, *Quantum and classical low-degree learning via a dimension-free Remez inequality*, arXiv:2301.01438v3 (2023). See the conclusion on PDF page 32.
- [2] J. Slote, A. Volberg, and H. Zhang, *Noncommutative Bohnenblust–Hille Inequality for qudit systems*, arXiv:2406.08509 (2024). See Theorem 7 on PDF pages 5–6 and its proof in Section 3.